

Score: 100%
25/25 points

WISH - Digital - Day-5

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Q1. What is the primary goal of logic synthesis?

- | | |
|--|--|
| ☐ A) Place cells on a die | ☐ C) Assign power rails to cells |
| ☒ B) Convert RTL (HDL) description into a gate-level netlist | ☐ D) Generate GDSII layout |

1/1 point

Q2. Which of the following is NOT a constraint file used during synthesis?

- | | |
|--|---|
| ☐ A) SDC (Synopsys Design Constraints) | ☒ C) DEF file |
| ☐ B) Liberty (.lib) file | ☐ D) UPF file |

1/1 point

Q3. What does timing-driven synthesis optimize?

- ☐ A) Power consumption only
- ☐ C) Number of metal layers
- ☒ **B) Setup and hold time violations in the design**
- ☐ D) Floorplan area

1/1 point

Q4. What is a "critical path" in synthesis?

- ☐ A) The shortest path in the design
- ☒ **C) The longest combinational path that determines maximum clock frequency**
- ☐ B) The path from power supply to ground
- ☐ D) The path used for scan testing

1/1 point

Q5. What does the Liberty (.lib) file contain?

- ☐ A) Placement coordinates of standard cells
- ☒ **C) Timing, power, and functional information of standard cells**
- ☐ B) Metal layer stack information
- ☐ D) Floorplan constraints

1/1 point

Q6. Which synthesis technique reduces power by eliminating unused logic?

- ☐ A) Technology mapping
- ☐ B) **Constant propagation and dead code elimination**
- ☐ C) Retiming
- ☐ D) Clock gating insertion

1/1 point

Q7. What is retiming in synthesis?

- ☐ A) Re-assigning clock frequencies
- ☐ B) **Redistributing registers across combinational logic to improve timing**
- ☐ C) Removing redundant flip-flops
- ☐ D) Recalculating setup times

1/1 point

Q8. What is the correct order of PnR steps?

- ☐ A) Routing → Placement → Floorplanning → CTS
- ☐ B) **Floorplanning → Placement → CTS → Routing**
- ☐ C) CTS → Floorplanning → Routing → Placement
- ☐ D) Placement → Floorplanning → Routing → CTS

1/1 point

Q9. What is the purpose of floorplanning?

- ☐ A) Assign clock frequencies to flip-flops
- ☐ B) **Define the die area, macro placement, and I/O pin positions**
- ☐ C) Generate timing reports
- ☐ D) Fix DRC violations

1/1 point

Q10. What does CTS stand for in VLSI design?

- ☐ A) Cell Timing Simulation
- ☐ B) **Clock Tree Synthesis**
- ☐ C) Critical Timing Setup
- ☐ D) Combinational Topology Synthesis

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Q11. What is the goal of Clock Tree Synthesis (CTS)?

- ☐ A) Minimize the number of clock buffers
- ☐ B) Maximize clock frequency
- ☐ C) Balance clock skew and ensure clock reaches all flip-flops with minimal latency difference
- ☐ D) **Route data signals**

1/1 point

Q12. What is "clock skew"?

- ☐ A) Difference in rise/fall times of a clock signal
- ☐ C) The clock frequency variation
- ☒ B) **Variation in clock arrival time at different flip-flops**
- ☐ D) Duty cycle distortion of the clock

1/1 point

Q13. What is the purpose of "filler cells" in physical design?

- ☐ A) Improve timing
- ☐ C) Buffer the clock signal
- ☒ B) **Provide power routing connectivity and satisfy density design rules**
- ☐ D) Act as scan chain elements

1/1 point

Q14. What is the primary function of an IC package?

- ☐ A) Increase transistor density
- ☐ C) Reduce leakage current
- ☒ B) **Protect the die, provide electrical connections, and dissipate heat**
- ☐ D) Improve clock frequency

1/1 point

15. What is "routing congestion"?

- ☐ A) Too many timing violations in a region
- ☐ B) **Insufficient routing resources to complete all required connections in an area**
- ☐ C) High power dissipation in a region
- ☐ D) Excessive IR drop in the power grid

1/1 point

16. What does DRC stand for?

- ☐ A) **Design Rule Check**
- ☐ B) Delay Reduction Constraint
- ☐ C) Dynamic Range Control
- ☐ D) Die Region Coverage

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17. What are the three main components of power dissipation in CMOS?

- ☐ A) Static, Leakage, and Short-circuit power
- ☐ B) Clock, Data, and I/O power
- ☐ C) **Leakage, Switching, and Internal**
- ☐ D) Thermal, Electrical, and Magnetic power

1/1 point

18. What technique is commonly used to reduce dynamic power?

- ☐ A) Increasing supply voltage
- ☐ B) **Clock gating**
- ☐ C) Increasing transistor size
- ☐ D) Using thicker metal layers

1/1 point

19. What is IR drop?

- ☐ A) **Voltage drop across metal interconnects due to resistive effects, reducing effective supply voltage**
- ☐ B) Current leakage through gate oxide
- ☐ C) Impedance mismatch at package pins
- ☐ D) Rise time degradation in logic gates

1/1 point

20. What does BGA stand for?

- ☐ A) Bonded Gate Array
- ☐ B) **Ball Grid Array**
- ☐ C) Buffer Gate Architecture
- ☐ D) Base Grid Alignment

1/1 point

Q21. A core area is $1\text{mm} \times 1\text{mm}$. Total utilization is 80% and Standard cell utilization is 75% . How much area (in mm^2) is available for routing and power grid?

- ☐ A) 0.75 mm^2 ☐ C) 0.20 mm^2
☒ B) **0.25 mm^2** ☐ D) 0.05 mm^2

1/1 point

22. A combinational path has gate delays of 100ps, 150ps, 120ps and wire delays of 30ps, 20ps between them. Clock period = 500ps, setup time = 60ps, clock skew = 0ps. What is the setup slack?

- ☒ A) **+20ps** ☐ C) +40ps
☐ B) +30ps ☐ D) -20ps

1/1 point

23. A flip-flop has: Clock-to-Q = 80ps, Combo delay = 300ps, Setup time = 60ps. What is the minimum clock period needed?

- ☐ A) 360ps ☒ C) **440ps**
☐ B) 380ps ☐ D) 300ps

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24. A path has setup slack = -40ps . The designer inserts a faster cell, reducing data path delay by 25ps . What is the new slack, and is the path still failing?

- ☒ **A) -15ps , still failing** ☐ C) -65ps , worse
- ☐ B) $+15\text{ps}$, now passing ☐ D) 0ps , borderline

1/1 point

25. A block runs at 1GHz , 1.0V . The designer reduces voltage to 0.9V (frequency unchanged). What is the % reduction in dynamic power?

- ☐ A) 10% ☐ C) 1%
- ☒ **B) 19%** ☐ D) 9%

1/1 point

Done



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